

INTRODUCTION TO HEALTH ECONOMICS

Quiz 1: Answer Key with Reasoning

Question 1

A person insists health care demand should be treated differently from demand for most everyday goods. What is the strongest reason behind this claim?

- a) The need for health care often arises suddenly, unlike planned purchases
- b) Health care providers change their prices more often than other sellers
- c) Health care tends to cost more per unit than everyday goods
- d) Health care is regulated by government agencies unlike most consumer goods

Answer: A

Reasoning: The need for health care can arise suddenly (an accident, a sudden illness) in a way that demand for most everyday goods cannot, since everyday purchases can usually be planned for. Options (b), (c), and (d) may be true of health care in some contexts, but none of them explain why health care demand specifically behaves differently from demand for ordinary goods; they describe surface features (pricing behavior, cost, regulation) rather than the core economic distinction, which is unpredictability of need.

Question 2

Rahul has full health insurance through his company. Over the next year, he starts skipping routine safety precautions at work, reasoning that any resulting medical costs are covered anyway. His insurer has no way of observing this change. Which term specifically describes this situation?

- a) Riskier individuals disproportionately choosing to buy coverage (Adverse Selection)
- b) One side possessing knowledge the other side lacks (Asymmetric Information)
- c) Behavior turning riskier once its cost is shared (Moral Hazard)
- d) A choice creating consequences for uninvolved third parties (Externality)

Answer: C (If someone answers B as well then don't deduct marks)

Reasoning: Rahul's behavior becomes riskier only after he is insured, and specifically because the cost of that riskier behavior is now shared with the insurer, this is the defining feature of moral hazard. Option (a), adverse selection, describes riskier people choosing to buy insurance in the first place, not a change in behavior after buying it. Option (b), asymmetric information, is a broader condition (one side knowing more than the other) that enables moral hazard but does not itself describe Rahul's behavioral change. Option (d), externality, would require Rahul's choice to impose a cost on someone outside the insurance contract, which is not what the scenario describes.

Question 3

An insurer offers a policy without medical screening, hoping to attract a broad mix of customers. Instead, mostly people who already expect to need expensive treatment sign up, while healthier people stay away. Which term specifically describes why the buyer pool ends up riskier?

- a) Behavior turning riskier once its cost is shared (Moral Hazard)
- b) A choice creating consequences for uninvolved third parties (Externality)
- c) Buyers reducing purchases sharply as the price rises (Price Elasticity)
- d) Riskier individuals disproportionately choosing to buy coverage (Adverse Selection)

Answer: D (If someone answers B as well then don't deduct marks)

Reasoning: The insurer ends up with a riskier pool of buyers specifically because the people most likely to need expensive treatment are also the most eager to buy the policy, this self-selection by risk type is adverse selection. Option (a), moral hazard, would require behavior to change after buying insurance, not the buyer pool being risky from the start. Option (b), externality, does not fit, since the effect described is on the insurer's own risk pool, not on outside third parties. Option (c), price elasticity, is irrelevant here since the scenario is about who chooses to buy, not how quantity demanded responds to price.

Question 4

A student with an untreated contagious illness continues attending crowded lectures, and several classmates who had no say in his decision fall sick as a result. Which term specifically describes the effect on the classmates?

- a) Behavior turning riskier once its cost is shared (Moral Hazard)
- b) A choice creating consequences for uninvolved third parties (Externality)
- c) Riskier individuals disproportionately entering into a transaction (Adverse Selection)
- d) Purchases falling noticeably as the price increases (Price Elasticity)

Answer: B

Reasoning: The classmates bear a cost (falling sick) as a direct result of a decision they had no part in, this is exactly what an externality is: a cost or benefit imposed on a third party outside the original decision. Option (a), moral hazard, does not apply because the student's behavior did not change due to being shielded from a cost by some contract. Option (c), adverse selection, is about riskier people opting into a transaction, which is unrelated to a contagious illness spreading involuntarily. Option (d), price elasticity, has no bearing on this scenario since no price or purchasing decision is involved.

Question 5

A health economist studies a particular treatment and finds its price elasticity of demand whose size is below 1. What does this specifically tell her about how patients respond to a price increase for that treatment?

- a) Patients reduce their use of the treatment by about as much as the price rise
- b) Patients reduce their use of the treatment only slightly compared to the price rise
- c) Patients actually increase their use of the treatment despite the price rise
- d) The size of patients' response cannot be judged from this number alone

Answer: B

Reasoning: When the absolute value of price elasticity of demand is below 1, demand is described as inelastic, meaning quantity demanded changes by a smaller percentage than price does. So patients cut back only slightly relative to the price increase. Option (a) describes elasticity equal to 1, which is not what "below 1" means. Option (c) would require a positive relationship between price and health care. On the contrary, quantity of health care demanded and prices have a negative relationship.. Option (d) is incorrect because the whole purpose of an elasticity figure is to quantify the size of the response; a value below 1 does tell us something specific and interpretable.

Question 6

A clinic raises the price of a purely cosmetic dental procedure, and monthly bookings fall sharply. A hospital raises the price of emergency insulin by a similar proportion, and monthly purchases barely change. Based on this pattern alone, what can be reasoned about the two treatments?

- a) Demand for the cosmetic procedure is more responsive to price than demand for insulin
- b) Demand for insulin is more responsive to price than demand for the cosmetic procedure
- c) Both treatments show an identical degree of responsiveness to changes in price
- d) Price changes have no measurable effect on demand for either treatment

Answer: A

Reasoning: Bookings for the cosmetic procedure fall sharply while insulin purchases barely move, even though both prices rose by a similar proportion. A larger percentage change in quantity for a similar percentage change in price means the cosmetic procedure's demand is more price responsive (more elastic). Option (b) reverses this comparison and is contradicted by the data given. Option (c) is wrong because the two treatments clearly respond very differently in the scenario. Option (d) is wrong because both treatments did show some response to price, insulin's response was simply much smaller, not zero.

Question 7

A researcher wants to know if having insurance increases how often people visit a doctor, so she compares people who already have insurance to those who don't, using existing data. Why can't this comparison reliably show that insurance itself causes more visits?

- a) Doctor visit records collected this way are generally unreliable and incomplete
- b) Insurance has already been conclusively shown to have no effect on visits
- c) Such comparisons rarely include a sufficient number of uninsured participants
- d) Insured and uninsured groups may differ in other ways that also affect visits

Answer: D

Reasoning: Because insurance status was not randomly assigned, insured and uninsured people may differ in other ways, income, health awareness, occupation, that independently affect how often they visit a doctor, so the raw comparison cannot isolate the effect of insurance alone. Option (a) is not the issue described; the flaw is about who is being compared, not the accuracy of the records. Option (b) is not something the scenario supports or establishes. Option (c) is a claim about sample size, which is a separate issue from the fundamental problem of comparing two potentially different kinds of people.

Question 8

To study whether insurance affects how much care people use, a researcher randomly places volunteers into different insurance plans by lottery, rather than letting them pick their own plan. What specific advantage does this give the study over simply observing people's existing choices?

- a) It guarantees every participant will use the same amount of care
- b) It makes the groups similar on average apart from the plan itself
- c) It removes any need to track outcomes over an extended period
- d) It ensures that only the healthiest participants are included in the study

Answer: B

Reasoning: Random assignment by lottery ensures that, on average, the people who end up in each insurance plan are similar to one another in every respect except which plan they were given, so any difference in outcomes can be more confidently attributed to the plan itself rather than to pre-existing differences between participants. Option (a) is not something randomization guarantees; people can still use different amounts of care within a plan. Option (c) is incorrect, since tracking outcomes over time is still necessary to observe any effect. Option (d) is false because random assignment does not select for health status; it deliberately avoids any selection at all.

Question 9

A study compares two groups: one that had to pay more for medical care, and one that paid less. The group paying more ended up using less care overall. Their general health turned out about the same as the other group, except for one thing: the poorest and sickest people in the higher-cost group saw their health get worse. What is the most reasonable way to interpret this?

- a) Higher medical costs directly improved health outcomes for the general population
- b) The amount of care people use has no real connection to its price
- c) Cutting back on care never affects health, even for the most vulnerable
- d) Most people cut back on low-value care, but vulnerable groups lost care that truly mattered

Answer: D

Reasoning: The overall similarity in health between the two groups suggests that most of the care people cut back on was not essential to their health, but the decline in health specifically among the poorest and sickest suggests that this group was cutting into care that actually mattered. Option (a) is contradicted by the scenario, since health did not improve, and even worsened for some. Option (b) is directly contradicted by the finding that the group paying more ended up using less care overall. Option (c) is directly contradicted by the finding that the poorest and sickest group's health did decline.

Question 10

A large-scale study found that even a sizeable increase in patients' out-of-pocket costs led to only a small fall in how much care they used overall. Based on this finding, which conclusion about health care demand is best supported?

- a) Health care demand is highly responsive to changes in its price
- b) Health care demand actually rises whenever its price also rises
- c) Health care demand is only mildly responsive to changes in its price
- d) Health care demand cannot be meaningfully measured using price changes

Answer: C

Reasoning: A sizeable increase in cost leading to only a small fall in usage is the definition of low price responsiveness, meaning demand is only mildly affected by price, consistent with inelastic demand. Option (a) describes the opposite pattern, a large fall in usage for a given price increase, which is not what the finding describes. Option (b) is inconsistent with the finding, since usage fell, not rose, as costs increased. Option (d) is incorrect because the study's finding is itself a meaningful measurement of how demand responds to price; it simply shows that response is modest.

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